

PRACTICE SHEET-1

1. If $n(A) = 115$, $n(B) = 326$, $n(A - B) = 47$, then what is $n(A \cup B)$ equal to?
 (1) 373 (2) 165
 (3) 370 (4) 394
2. If $P(A)$ denotes the power set of A and A is the void set, then what is number of elements in $P\{P\{P(A)\}\}$?
 (1) 0 (2) 1
 (3) 4 (4) 16
3. If $N_a = \{ax \mid x \in N\}$, then what is $N_{12} \cap N_8$ equal to?
 (1) N_{12} (2) N_{20}
 (3) N_{24} (4) N_{48}
4. If $X = \{4^n - 3n - 1 \mid n \in N\}$ and $Y = \{9(n - 1) \mid n \in N\}$, then $X \cup Y$ equals to
 (1) X (2) Y
 (3) N (4) A null set
5. What is the number of proper subsets of a given finite set with n elements?
 (1) $2n - 1$ (2) $2n - 2$
 (3) $2^n - 1$ (4) $2^n - 2$
6. If A , B and C are three finite sets, then $[(A \cup B) \cap C]'$ equals to
 (1) $A' \cup B' \cap C'$ (2) $A' \cap B' \cap C'$
 (3) $A' \cap B' \cup C'$ (4) $A \cap B \cap C$
7. If A and B are subsets of a set X , then what is the value of $\{A \cap (X - B)\} \cup B$?
 (1) $A \cup B$ (2) $A \cap B$
 (3) A (4) B
8. Which one of the following is correct?
 (1) $A \cup P(A) = P(A)$
 (2) $A \cap P(A) = A$
 (3) $A - P(A) = A$
 (4) $P(A) - \{A\} = P(A)$
 Here, $P(A)$ denotes the power set of a set A .
9. If A , B and C are three sets, then $A - (B - C)$ equals to
 (a) $A - (B \cap C)$ (b) $(A - B) \cup C$
 (c) $(A - B) \cup (A \cap C)$ (d) $(A - B) \cup (A - C)$
10. Let $S = \{2, 4, 6, \dots, 20\}$, what is the number of subsets does S have?
 (a) 10 (b) 20
 (c) 512 (d) 1024
11. If A , B and C are subsets of a given set, then which one of the following is relations is
 (a) $A \cup (A \cap B) = A \cup B$
 (b) $A \cap (A \cup B) = A$
 (c) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
 (d) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
12. If a set A contains 3 elements and another set B contains 6 elements, then what is the minimum number of elements that $(A \cup B)$ can have?
 (a) 3 (b) 6
 (c) 8 (d) 9
13. If $A = \{x: 0 \leq x \leq 2\}$ and $B = \{y: y \text{ is a prime number}\}$, then what is $A \cap B$ equal to?
 (a) $\emptyset$ (b) $\{1\}$
 (c) $\{2\}$ (d) $\{1, 2\}$
14. If $A = \{x : x \text{ is a multiple of } 2\}$, $B = \{x : x \text{ is a multiple of } 5\}$ and $C = \{x : x \text{ is a multiple of } 10\}$, then $A \cap \{B \cap C\}$ is equal to:
 (a) A
 (b) B
 (c) C
 (d) $\{x : x \text{ is a multiple of } 100\}$
15. If $S = \{x : x^2 + 1 = 0, x \text{ is real}\}$, then S is:
 (a) $\{-1\}$ (b) $\{0\}$
 (c) $\{1\}$ (d) An empty set
16. Consider the following in respect of sets A and B :
 1. $(A - B) \cup B = A$
 2. $(A - B) \cup A = A$
 3. $(A - B) \cap B = \phi$
 4. $A \subseteq B \Rightarrow A \cup B = B$
 Which of the above are correct?
 (1) 1, 2 and 3 (2) 2, 3 and 4
 (3) 1, 3 and 4 (4) 1, 2 and 4
17. Let S be a set of all distinct numbers of the form $\frac{p}{q}$ where $p, q \in (1, 2, 3, 4, 5, 6)$. What is the cardinality of the set S ?
 (1) 21 (2) 23
 (3) 32 (d) 36
18. If $A = \{x : x \text{ is a multiple of } 3\}$ and $B = \{x : x \text{ is a multiple of } 4\}$ and $C = \{x : x \text{ is a multiple of } 12\}$

- of 12}, then which one of the following is null set?
- (1) $(A \setminus B) \cup C$ (2) $(A \setminus B) \setminus C$
(3) $(A \cap B) \cap C$ (4) $(A \cap B) \setminus C$
19. Which one of the following is a null set?
(a) $\{0\}$
(b) $\{\{\}\}$
(c) $\{\{\}\}$
(d) $\{x \mid x^2 + 1 = 0, x \in \mathbb{R}\}$
20. If $A = \{1, 3, 5, 7\}$, then what is the cardinality of the power set $P(A)$?
(1) 8 (2) 15
(3) 16 (4) 17
21. If A and B are any two sets, then what is the value of $A \cap (A \cup B)$?
(a) Complement of A
(b) Complement of B
(c) B
(d) A
22. Let $A = \{x : x \text{ is a square of a natural number and } x \text{ is less than } 100\}$ and B is a set of even natural numbers. What is the cardinality of $A \cap B$?
(a) 4 (b) 5
(c) 9 (d) None of these
23. Let $U = \{x \in \mathbb{N} : 1 \leq x \leq 10\}$ be the universal set, N being the set of natural numbers. If $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 6, 10\}$, then what is the complement of $(A - B)$?
(a) $\{6, 10\}$
(b) $\{1, 4\}$
(c) $\{2, 3, 5, 6, 7, 8, 9, 10\}$
(d) $\{5, 6, 7, 8, 9, 10\}$
24. Let $A = \{x : x \text{ is a square of a natural number and } x \text{ is less than } 100\}$ and B is set of even natural numbers. What is the cardinality of $A \cap B$?
(a) 4 (b) 5
(c) 9 (d) None of these
25. Let $U = \{x \in \mathbb{N} : 1 \leq x \leq 10\}$ be the universal set, N being the set of natural numbers. If $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 6, 10\}$, then what is the complement of $(A - B)$?
(a) $\{6, 10\}$
(b) $\{1, 4\}$
(c) $\{2, 3, 5, 6, 7, 8, 9, 10\}$
(d) $\{5, 6, 7, 8, 9, 10\}$
26. For a set A , consider the following statements
I. $A \cup P(A) = P(A)$
II. $\{A\} \cap P(A) = A$
III. $P(A) - \{A\} = P(A)$
Where P denotes power set.
Which of the statements given above is/are not correct?
(1) I only (2) II only
(3) III only (4) I, II and III
27. If $A = \{a, b, c\}$, then what is the number of proper subsets of A ?
(a) 5 (b) 6
(c) 7 (d) 8
28. If $A = \{4n + 2 \mid n \text{ is natural number}\}$ and $B = \{3n \mid n \text{ is a natural number}\}$, then what is $(A \cap B)$ equal to?
(1) $\{12n^2 + 6n \mid n \text{ is a natural number}\}$
(2) $\{24n - 12 \mid n \text{ is a natural number}\}$
(3) $\{12n^2 - 6n \mid n \text{ is a natural number}\}$
(4) $\{12n - 6 \mid n \text{ is a natural number}\}$
29. If the cardinality of a set A is 4 and that of a set B is 3, then what is the cardinality of the set $A \Delta B$?
(a) 1
(b) 5
(c) 7
(d) Cannot be determined as the sets A and B are not given.
30. If $A = P\{1, 2\}$, where P denotes the power set, then which one of the following is correct?
(a) $\{1, 2\} \subset A$ (b) $1 \in A$
(c) $\phi \notin A$ (d) $\{1, 2\} \in A$

ANSWER KEY

1.	(a)	2.	(d)	3.	(c)	4.	(b)	5.	(c)	6.	(c)	7.	(a)	8.	(c)	9.	(c)	10.	(d)
11.	(b)	12.	(b)	13.	(c)	14.	(c)	15.	(d)	16.	(b)	17.	(b)	18.	(d)	19.	(d)	20.	(c)
21.	(d)	22.	(a)	23.	(c)	24.	(a)	25.	(c)	26.	(d)	27.	(c)	28.	(d)	29.	(d)	30.	(d)

PRACTICE SHEET-2

1. In an examination 45% of the students pass in English, 35% pass in Hindi and 20% pass in both. What percent of students fail in both English and Hindi?
(a) 10% (b) 20%
(c) 25% (d) 40%
 2. In an examination 60% of the students pass in English, 70% pass in Hindi and 40% pass in both. What percent of students fail in both English and Hindi?
(a) 10% (b) 20%
(c) 25% (d) 30%
 3. In an examination 70% of the candidates passed in English 80% passed in Mathematics 10% failed in both the subjects. If 144 candidates passed in both, the total number of candidates were:
(a) 125 (b) 200
(c) 240 (d) 375
 4. In an examination, 52% students failed in Hindi and 42% in English. If 17% failed in both the subjects, what percentage of students passed in both the subjects?
(a) 38% (b) 33%
(c) 23% (d) 18%
 5. In a group of students, 70% can speak English and 65% can speak Hindi. If 27% of the students can speak none of the two languages, then what percent of the group can speak both the languages.
(a) 38% (b) 62%
(c) 28% (d) 23%
 6. In an examination, 35% of the candidates failed in Mathematics and 25% in English. If 10% failed in both Mathematics and English, then how much percent of candidates passed in both the subjects?
(a) 50% (b) 55%
(c) 57% (d) 60%
 7. In an examination, 93% of students passed and 259 failed. The total number of students appearing at the examination was
(a) 3700 (b) 3850
(c) 3950 (d) 4200
 8. 90% of the students in a school passed in English, 85% passed in Mathematics and 150 students passed in both the subjects. If no student failed in both the subjects, find the total number of students.
(a) 120 (b) 220
(c) 200 (d) 300
 9. In an examination 75% candidates passed in English and 60% passed in Mathematics. 25% failed in both and 240 passed the examination. Find the total number of candidates.
(a) 492 (b) 300
(c) 500 (d) 400
 10. In an examination, 35% of total students failed in Hindi, 45% failed in English and 20% failed in both. Find the percentage of those students who passed in both the subjects?
(a) 45% (b) 35%
(c) 20% (d) 40%
- Direction (for next five):** The Students of a class are offered three languages (Hindi, English and French). 15 students learn all the three language whereas 28 students do not learn any language. The number of students learning Hindi and English but not French is twice the number of students learning Hindi and French but not English. The number of students learning English and French but not Hindi is thrice the number of students learning Hindi and French but not English. 23 students learn only Hindi and 17 students learn only English. The total number of students learning French is 46 and the total number of students learning only French is 11.
11. How many students learn precisely two languages?
(a) 55 (b) 40
(c) 30 (d) 13
 12. How many students learn at least two language?
(a) 15 (b) 30
(c) 45 (d) 55

13. What is the total strength of the class?

- (a) 124 (b) 100
(c) 96 (d) 66

14. How many students learn English and French?

- (a) 30 (b) 43
(c) 45 (d) 73

15. How many students learn at least one language?

- (a) 45 (b) 51
(c) 96 (d) None of these

Direction (for next three): Read the following information carefully to answer the questions that follow.

In a survey of 25 students, it was found that 15 have taken Mathematics, 12 have taken Physics and 11 taken Chemistry, 5 have taken Mathematics and Chemistry, 9 have taken Mathematics and Physics, 4 have taken Physics and Chemistry and 3 have taken all the three subjects.

16. The number of students who have taken only Physics, is

- (a) 2 (b) 3 (c) 5 (d) 6

17. The number of students who have taken only two subjects, is

- (a) 7 (b) 8
(c) 9 (d) 10

18. Consider the following statements

I. The number of students who have taken only one subject is equal to the number of students who have taken only two subjects.
II. The number of students who have taken at least two subjects is four times the number of students who have taken all the three subjects.

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Direction for (next two): In a school, all the students play at least one of the games - chess, carom and table tennis. 60 play chess, 50 play table tennis, 48 play carom, 12 play chess and carom, 15 play carom and table tennis, 20 play table tennis and chess.

19. What can be the minimum number of students in the school?

- (a) 123 (b) 111
(c) 95 (d) 63

20. What can be the maximum number of students in the school?

- (a) 111 (b) 123
(c) 125 (d) 135

ANSWER KEY

1.	(d)	2.	(a)	3.	(c)	4.	(c)	5.	(b)	6.	(a)	7.	(a)	8.	(c)	9.	(d)	10.	(d)
11.	(c)	12.	(c)	13.	(a)	14.	(a)	15.	(c)	16.	(a)	17.	(c)	18.	(b)	19.	(b)	20.	(b)

CDS PYQ

1. In a school there are 30 teachers who teach Mathematics or Physics. Of these teachers, 20 teach Mathematics and 15 teach Physics, 5 teach both Mathematics and Physics. The number of teachers teaching only Mathematics is
[CDS 2013(I)]
(a) 5 (b) 10
(c) 15 (d) 20
2. If $A = \{x : x \text{ is an odd integer}\}$ and $B = \{x : x^2 - 8x + 15 = 0\}$. Then, which one of the following is correct?
[CDS 2013(I)]
(a) $A = B$ (b) $A \subseteq B$
(c) $B \subseteq A$ (d) $A \subseteq B^c$
3. Consider the following in respect of the sets A and B .
I. $(A \cap B) \subseteq A$
II. $(A \cap B) \subseteq B$
III. $A \subseteq (A \cup B)$
Which of the above are correct?
[CDS 2013(I)]
(a) I and II (b) II and III
(c) I and III (d) I, II and III
4. The set of natural numbers is closed under
I. addition
II. subtraction
III. multiplication
IV. division
Which of the above is/are correct?
[CDS 2013(I)]
(a) Only I (b) Both I and III
(c) I, II and III (d) Both III and IV
5. In a class of 110 students, x students take both Mathematics and Statistics, $2x + 20$ students take Mathematics and $2x + 30$ students take Statistics. There are no students who take neither Mathematics nor Statistics. What is x equal to?
[CDS 2013(II)]
(a) 15 (b) 20
(c) 25 (d) 30
6. Out of 105 students taking an examination English and Mathematics, 80 students pass in English, 75 students pass in Mathematics 10 students fail in both the subjects. How many students fail in only one subject?
[CDS 2014(I)]
(a) 26 (b) 30
(c) 35 (d) 45
7. If A and B are any two non-empty subsets of a set E , then what is $A \cup (A \cap B)$ equal to?
[CDS 2014(I)]
(a) $A \cap B$ (b) $A \cup B$
(c) A (d) B
8. If A is a non-empty subset of a set E , then what is $E \cup (A \cap \phi) - (A - \phi)$ equal to?
[CDS 2014(I)]
(a) A
(b) Complement of A
(c) ϕ
(d) E
9. Let A and B be finite non-empty sets with the number of elements in $A = m$ and number of elements in $B = n$. Let $m > n$. If for some integer $k \geq 1$, the number of non-empty subsets of $A = 2^k +$ the number of non-empty subsets of B , then which one of the following is correct?
[CDS 2015(I)]
(a) $m = n + 2$
(b) $m = n + 1$
(c) $m = n + p$ for some odd prime number p
(d) $m = n + t$ for some composite number t
10. Let A denote the set of quadrilaterals having two diagonals equal and bisecting each other. Let B denote the set of quadrilaterals having diagonals bisecting each other at 90° . Then $A \cap B$ denotes
[CDS 2015(II)]
(a) the set of parallelograms
(b) the set of rhombuses
(c) the set of squares
(d) the set of rectangles
11. In a gathering of 100 people, 70 of them can speak Hindi, 60 can speak English and 30 can speak French. Further, 30 of them can speak both Hindi and English. 20 can speak both Hindi and French. If x is the number of people who can speak both English and French, then which one of the following is correct? (Assume that everyone can speak at least one of the three languages)
[CDS 2016(I)]

- (a) $9 < x \leq 30$ (b) $0 \leq x < 8$
(c) $x = 9$ (d) $x = 8$

- 12.** A is a set of positive integers such that when divided by 2, 3, 4, 5 and 6 leaves the remainder 1, 2, 3, 4 and 5 respectively. How many integers between 0 and 100 belong to the set A?

[CDS 2016(II)]

- (a) No integer (b) One
(c) Two (d) Three

- 13.** In the quadratic equation $x^2 + ax + b = 0$, a and b can take any value from the set {1, 2, 3, 4}. How many pairs of values of a and b are possible in order that the quadratic equation has real roots?

[CDS 2016(II)]

- (a) 6 (b) 7
(c) 8 (d) 16

- 14.** Let $A = \{7, 8, 9, 10, 11, 12\}$ and $B = \{7, 10, 14, 15\}$. What is the number of elements in $(A-B)$ and $(B-A)$ respectively?

[CDS 2016(II)]

- (a) 2 and 4 (b) 4 and 2
(c) 2 and 2 (d) 4 and 4

- 15.** Let S be a set of first ten natural numbers. What is the possible number of pairs (a, b) where $a, b \in S$ and $a \neq b$ such that the product ab (>12) leaves remainder 4 when divided by 12?

[CDS 2016(II)]

- (a) 4 (b) 6
(c) 8 (d) 10

- 16.** If $A = \{x : x \text{ is a multiple of } 7\}$, $B = \{x : x \text{ is a multiple of } 5\}$ and $C = \{x : x \text{ is a multiple of } 35\}$, then which one of the following is a null set?

[CDS 2017(I)]

- (a) $(A - B) \cup C$ (b) $(A - B) - C$
(c) $(A \cap B) \cap C$ (d) $(A \cap B) - C$

- 17.** Which one of the following is a correct statement?

[CDS 2017(II)]

- (a) $\{x : x + 5 = 5\} = \phi$
(b) $\{x : x + 5 = 5\} = \{0\}$
(c) $\{x : x + 5 = 5\} = 0$
(d) $\{x : x + 5 = 5\} = \{\phi\}$

- 18.** In an examination, 35% students failed in Hindi, 45% students failed in English and 20% students failed in both the subjects.

What is the percentage of students passing in both the subjects?

[CDS 2017(II)]

- (a) 0 (b) 20
(c) 30 (d) 40

- 19.** In an examination 50% of candidates failed in English, 40% failed in Hindi and 15% failed in both the subjects. The percentage of candidates who passed in both English and Hindi is:

[CDS 2018(II)]

- (a) 20% (b) 25%
(c) 60% (d) 75%

- 20.** Out of 85 children playing badminton or table tennis or both, the total number of girls in the group is 70% of the total number of boys in the group. The number of boys playing only badminton is 50% of the number of boys and the total number of boys playing badminton is 60% of the total number of boys. The number of children playing only table tennis is 40% of the total number of children and a total of 12 children play badminton and table tennis both. The number of girls playing only badminton is

[CDS 2018(II)]

- (a) 14 (b) 16
(c) 17 (d) 35

- 21.** In a class of 60 boys, there are 45 boys who play chess and 30 boys who play carom. If every boy of the class plays at least one of the two games, then how many boys play carom only?

[CDS 2018(II)]

- (a) 30 (b) 20
(c) 15 (d) 10

Consider the following for the next 04 (four) items that follow:

In an examination of Class XII, 55% students passed in Biology, 62% passed in Physics, 60% passed in Chemistry, 25% passed in Physics and Biology, 30% passed in Physics and Chemistry, 28% passed in Biology and Chemistry. Only 2% failed in all the subjects.

[CDS 2018(II)]

- 22.** What percentage of students passed in all the three subjects?

- (a) 6 (b) 5
(c) 4 (d) 3

- 23.** What percentage of students passed in exactly one subject?
(a) 21 (b) 23
(c) 25 (d) 27
- 24.** If the number of students is 360, then how many passed in at least two subjects?
(a) 270 (b) 263
(c) 265 (d) 260
- 25.** What is the ratio of number of students who passed in both Physics and Chemistry to number of students who passed in both Biology and Physics but not Chemistry?
(a) 7:10 (b) 10:7
(c) 9:7 (d) 7:9
- 26.** In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% failed in both the subjects, then what percent passed in both subjects?
[CDS 2019(I)]
(a) 77 (b) 58
(c) 48 (d) 23
- 27.** If $X = \{a, \{b\}, c\}$, $Y = \{\{a\}, b, c\}$ and $Z = \{a, b, \{c\}\}$, then $(X \cap Y) \cap Z$ equals to
[CDS 2019(II)]
(a) $\{a, b, c\}$ (b) $\{\{a\}, \{b\}, \{c\}\}$
(c) $\{\phi\}$ (d) ϕ
- 28.** In a competitive examination, 250 students have registered. Out of these, 50 students have registered for Physics, 75 students for Mathematics and 35 students for both Mathematics and Physics. What is the number of students who have registered neither for Physics nor for Mathematics?
[CDS 2020(I)]
(a) 90 (b) 100
(c) 150 (d) 160
- Direction: Consider the following for the four (04) items that follow:**
500 candidates appeared in an examination comprising tests in English, Hindi and Mathematics. 30 candidates failed in English only; 75 failed in Hindi only; 50 failed in Mathematics only; 15 failed in both English and Hindi; 17 failed in both Hindi and Mathematics; 17 failed in both Mathematics, English; 5 failed in all three tests.
- 29.** What is the percentage of candidates who failed in at least two subjects?
[CDS 2022(I)]
(a) 5.4% (b) 6.4%
(c) 6.8% (d) 7.8%
- 30.** What is the percentage of candidates who failed in only one subject?
[CDS 2022(I)]
(a) 28% (b) 31%
(c) 35.8% (d) 38.8%
- 31.** What is the percentage of candidates who failed in at least one subject?
[CDS 2022(I)]
(a) 31% (b) 35.4%
(c) 38.8% (d) 41.5%
- 32.** How many candidates passed in two or more subjects?
[CDS 2022(I)]
(a) 461 (b) 405
(c) 345 (d) 306
- 33.** Let $X = \{x | x = 2 + 4k, \text{ where } k = 0, 1, 2, 3, \dots, 24\}$. Let S be a subset of X such that the sum of no two elements of S is 100. What is the maximum possible number of elements in S ?
[CDS - 2023 (1)]
(a) 10 (b) 11
(c) 12 (d) 13
- 34.** In a party of 150 persons, 75 persons take tea, 60 persons take coffee and 50 persons take milk. 15 of them take both tea and coffee, but no one taking milk takes tea. If each person in the party takes at least one drink, then what is the number of persons taking milk only.
[CDS - 2023 (1)]
(a) 50 (b) 40
(c) 30 (d) 20

ANSWER KEY

1.	(c)	2.	(c)	3.	(d)	4.	(b)	5.	(b)	6.	(c)	7.	(c)	8.	(b)	9.	(b)	10.	(c)
11.	(a)	12.	(b)	13.	(b)	14.	(b)	15.	(c)	16.	(d)	17.	(b)	18.	(d)	19.	(b)	20.	(a)
21.	(c)	22.	(c)	23.	(b)	24.	(a)	25.	(b)	26.	(d)	27.	(d)	28.	(d)	29.	(d)	30.	(b)
31.	(c)	32.	(a)	33.	d	34.	c												